System Design Workshop: End-to-End Enterprise RAG System

Scaler School of Technology | ML System Design and LLMOps | Week 7 — Session 1/2 | Student Worksheet

The ML System Design Rubric — Your Framework for Today

Use this 10-step framework to design the system. Work through the steps in order on pages 2–3.

1. **Clarify the Problem** — scope, constraints, success metrics
2. **Context** — business objective, users, cost of failure
3. **Problem Statement** — input, output, what to predict/retrieve
4. **Define Requirements** — functional and non-functional
5. **Estimations** — corpus size, QPS, storage, LLM cost/day
6. **System Architecture** — (a) Data Collection • (b) Data Engineering • (c) Feature Engineering • (d) Preprocessing • (e) Models • (f) Post-processing
7. **Evaluation** — offline metrics, online north-star, guardrails
8. **A/B Testing** — unit, split, duration, biases to flag
9. **Deployment & Monitoring** — pipelines, rollout, dashboards
10. **Risk & Trade-offs** — RBAC approach, LLM provider, chunking, hallucination, deletion

The Problem

Design a RAG system for an enterprise internal knowledge base.

Scale: 10,000 employees, 500,000 documents (Confluence, Jira, Slack, Google Drive, SharePoint).

Requirements: p99 latency < 3s (LLM included) • role-based access control (employees only see authorised docs) • citations traceable to source • new docs indexed within 30 minutes • GDPR right-to-deletion • 500 QPS

Steps 1–3 — Clarify, Context, Problem Statement

Ask your clarifying questions, note the context, then write a precise problem statement.

Clarifying questions I would ask:

Context (business objective, users, cost of a wrong answer):

Problem statement (input, output, constraint):

Step 4 — Requirements

Requirement	Value / assumption	Implication for design
Latency SLA		
Throughput (QPS)		
RBAC correctness		
Freshness SLA		
GDPR deletion		

Step 5 — Estimations

What	Your estimate	Design implication
Number of chunks (500K docs)		
Dense index size		
LLM cost per day		
Ingestion rate needed		

Step 6 — System Architecture

Label each component: what it does, key tools, key decisions. CRITICAL: where does RBAC enforcement happen?

(a–b) Data Collection & Engineering

(c–d) Feature Engineering & Retrieval Pipeline

(e–f) Models & Post-processing + Full System Diagram

Draw components as labelled boxes. Show data flow with arrows. Mark where RBAC filter applies.

Step 7 — Evaluation

Type	Metric(s)	Why this metric?
Offline		
Online north-star		
Guardrail		

Step 8 — A/B Testing

Unit of randomisation, split, duration, and the specific enterprise bias you must watch for.

Step 9 — Deployment and Monitoring

Deployment (ingestion + serving + deletion pipeline):

6 metrics to monitor:

1.
2.
3.
4.
5.
6.

Step 10 — Risk Management and Trade-offs

Trade-off	Your choice	Reasoning
RBAC pre-filter vs. post-filter		
External LLM vs. self-hosted		
Shared vs. per-source embedding model		
Chunk size (128t vs. 512t vs. 1024t)		
Cache strategy for LLM answers		

What Breaks First When a Major Re-org Happens?

A department merges: 50,000 documents change access permissions overnight. What breaks and how do you handle it?

Name: _____ Group: _____ Date: _____